

Review of Probability

Probability:

Probability is a measure of certainty (surety) of an event. It takes a value between '0' and '1'. For an impossible event, the probability is zero. For a sure event, the probability is one.

Example: If a regular (fair) coin is tossed, the probability that a 'head' occurs is $1/2 = 0.50$ (50% chance).

Example: If a fair dice is tossed, the probability that a '3' occurs is $1/6 = 0.167$ (16.7% chance).

Sample space:

The set of all possible outcomes of a random experiment is called the sample space. It is usually denoted by S . It is comparable to the universal set in set theory.

Example: In coin tossing experiment, $S = \{H, T\}$.

Example: In dice throwing experiment, $S = \{1, 2, 3, 4, 5, 6\}$.

Example: In a cricket match, $S = \{\text{win, loss, tie, postponed, cancelled}\}$.

Event:

Any subset of the sample space is called an event. When a random experiment is going to be conducted, we are interested in the probabilities of different events.

Example: In dice throwing, $A = \{1\}$, $B = \{1, 4, 5\}$, etc. are events. Note that B denotes the event that 1 or 4 or 5 occurs. They cannot happen together!

Example: In cricket match, $C = \{\text{loss}\}$, $D = \{\text{win, tie}\}$, etc. are events.

Events are sets. So, upper-case letters should be used for notation.

Operations on events:

$A \cup B$ occurs when A or B (or both) occur.

$A \cap B$ occurs when both A and B occur.

A^c occurs when A does not occur.

Classical or mathematical definition of probability:

Let a random experiment have n possible outcomes that are mutually exclusive, exhaustive and equally likely (all the outcomes have same chance). If m of these outcomes are favorable to an event A , then probability of A is given by:

$$P(A) = \frac{m}{n}$$

Note that, we cannot use this formula when all the outcomes are not equally likely, or the total number of possible outcomes, n , is infinite.

Example: Let a **fair** dice be thrown. Let $A = \{1, 4, 5\}$. Then

$$P(A) = \frac{m}{n} = \frac{3}{6} = 0.5.$$

Example: Consider a cricket match. $S = \{\text{win, loss, tie, postponed, cancelled}\}$. Let $D = \{\text{win, tie}\}$. Then, we should NOT say $P(D) = \frac{2}{5}$. (Why not?)

Empirical or statistical or frequency definition of probability

Empirical means observation-based, i.e., data-based.

Let an experiment be conducted n times, where n is large. If an event A occurs f_A times, then

$$P(A) \approx \frac{f_A}{n}$$

We cannot use this if the experiment cannot be repeated under the same conditions.

Example: An **unfair** dice is tossed 1000 times. “6” occurred 400 times, Then,

$$P(6) \approx \frac{400}{1000} = 0.40$$

Axioms of probability

Probability follows the following three axioms:

1. $0 \leq P(A) \leq 1$.
2. $P(S) = 1$.

3. When A and B are mutually exclusive events, i.e., $A \cap B = \emptyset$, then

$$P(A \cup B) = P(A) + P(B).$$

Theorem 1: $P(\emptyset) = 0$.

Theorem 2: $P(A^c) = 1 - P(A)$

Theorem 3 (Addition Law): For any two sets A and B , we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Exercise:

In a community, 25% of the families have cars, 15% have washing machines and 10% have both. A family is selected at random from the community. What is the probability that the family has (i) a car or a washing machine? (ii) neither a car nor a washing machine?

Solution:

$$P(C) = 0.25$$

$$P(W) = 0.15$$

$$P(C \cap W) = 0.10$$

$$(i) P(C \cup W) = 0.25 + 0.15 - 0.10 = 0.30$$

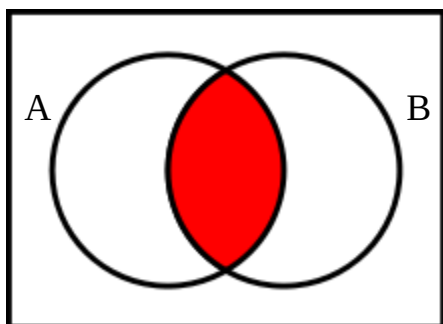
$$(ii) P((C \cup W)') = 1 - P(C \cup W) = 1 - 0.30 = 0.70$$

Conditional Probability

Probability calculated under a condition or additional information is called conditional probability.

Notation

$P(B | A)$ is the conditional probability that event B occurs given that the event A occurs.



The formula for conditional probability:

$$P(B | A) = \frac{P(A \cap B)}{P(A)}, \text{ when } P(A) \neq 0$$

Exercise

In a class, 60% of the students are boys. 15% of the students are boys that play cricket. A student is selected at random. If the selected student is a boy, what is the probability that he plays cricket?

Solution

$$P(B) = 0.60$$

$$P(B \cap C) = 0.15$$

$$P(C | B) = \frac{P(B \cap C)}{P(B)} = \frac{0.15}{0.60} = 0.25$$

If we cross-multiply the conditional probability formula, we get the Multiplication Rule:

$$P(A \cap B) = P(A) P(B | A)$$

Exercise

In a class, 60% of the students are boys. 25% of the boys play cricket. A student is selected at random. What is the probability that the student will be a boy who plays cricket?

Solution

$$P(B) = 0.60$$

$$P(C | B) = 0.25$$

$$P(B \cap C) = P(B) P(C | B) = 0.60 \times 0.25 = 0.15$$

Independence of events

Two events A and B are independent if occurrence of A does not change the probability of B , that is,

$$P(B | A) = P(B).$$

If A and B are independent, then the multiplication rule reduces to

$$P(A \cap B) = P(A) P(B)$$

If A , B and C are independent, then

$$P(A \cap B \cap C) = P(A) P(B) P(C)$$

Exercise

In a town, 20% of families have cars, 10% of families have washing machines and 5% of families have both. Are “having a car” and “having a washing machine” independent?

Solution

$$P(C) = 0.20, P(W) = 0.10 \text{ and } P(C \cap W) = 0.05.$$

We see that, $0.20 \times 0.10 = 0.02 \neq 0.05$. That is, $P(C) P(W) \neq P(C \cap W)$.

Therefore, C and W are not independent.

Example

Three fair coins are tossed. $S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$. Each outcome has probability $1/8$. We can use independence to calculate the probabilities.

$$P(HHH) = P(H) P(H) P(H) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$$

Exercise

An unfair coin (with probability of head 0.7) is tossed three times. Calculate the probabilities of all the possible outcomes.

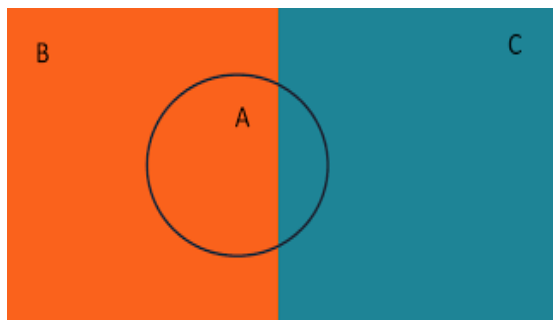
Solution: Do it yourself

Exercise

A machine works if at least two of the three components A, B and C work. Each component works independently with probability 0.8. What is the probability that the machine works?

Solution: Do it yourself

Bayes' Theorem



Statement:

Let B and C be two partitions of S . That is, $B \cup C = S$ and $B \cap C = \emptyset$. Let A be any event. Let $P(B)$, $P(C)$, $P(A | B)$ and $P(A | C)$ be known. Then

$$(i) \quad P(A) = P(B) P(A | B) + P(C) P(A | C)$$

$$(ii) \quad P(B | A) = \frac{P(B) P(A | B)}{P(A)}$$

Proof:

(i)

$$A = (B \cap A) \cup (C \cap A)$$

$$\therefore P(A) = P(B \cap A) + P(C \cap A)$$

$$\therefore P(A) = P(B) P(A | B) + P(C) P(A | C)$$

(ii)

$$P(B | A) = \frac{P(B \cap A)}{P(A)}$$

$$\therefore P(B | A) = \frac{P(B) P(A | B)}{P(A)}$$

Example (Quality of a blood test):

A blood test gives 97% positive results among the diseased people, and 2% positive results among the healthy people.

In a community, 1% of the people have that particular disease. A person is selected at random from the community and the blood test is performed on that person.

(i) What is the probability that the test result will be positive?

(ii) If the test is positive, what is the probability that the person has the disease?

Solution:

D : the selected person has the disease.

H : the selected person is healthy.

$+$: the test result of the person is positive.

(Set notation rule is ignored for comfort.)

$$P(D) = 0.01, P(+ | D) = 0.97$$

$$P(H) = 0.99, P(+ | H) = 0.02$$

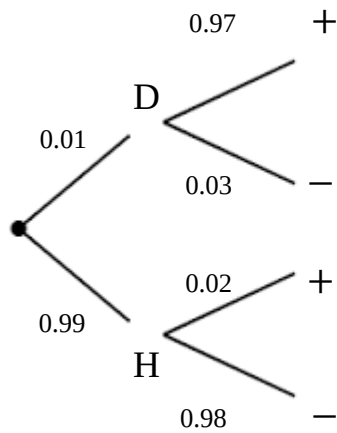
(i)

$$P(+) = 0.01 \times 0.97 + 0.99 \times 0.02 = 0.0295$$

(ii)

$$P(D | +) = \frac{P(D) P(+ | D)}{P(+)} = \frac{0.01 \times 0.97}{0.0295} = 0.33$$

Solution using a tree:



From the above tree,

$$P(+) = 0.01 \times 0.97 + 0.99 \times 0.02 = 0.0295$$

$$P(D | +) = \frac{P(D) P(+ | D)}{P(+)} = \frac{0.01 \times 0.97}{0.0295} = 0.33$$

Note:

1. If the test result is positive, there is only 33% chance that the person has the disease, and 67% chance that the person is healthy!! Why? [Hint: Healthy people are creating the problem. 😊]
2. $P(D)$ is called the '**prior**' probability, that is, probability before getting 'data' (test result). $P(D | +)$ is called the '**posterior**' probability, that is, probability after getting data.
3. What is the contribution of the blood test? Before getting the test result, we could say that there is only 1% chance (prior probability 0.01) that the person has the disease. If the test result is positive, we can say that there is a 33% chance (posterior probability 0.33) that the person has the disease. From 1% to 33% - this is the contribution of the blood test. Unfortunately, this is not enough.
4. How to develop a useful blood test? Make sure that there are no 'false positives' (healthy people testing positive). If we cannot do that, 'false positive' cases will dominate 'true positive' cases (because, there are so many healthy people), and doctors will have nightmares.

* Bayes' Rule is often used for **classification problems**. In the above example, it is used for classifying a person as diseased or healthy based on test result (data).